Radar Search Pipeline AS IS

Current implementation guide for Radar candidate and signal search

Status: AS IS

Product area: Radar candidate discovery pipeline

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Canonical source: current implementation, tests, ROADMAP.md, and Radar run diagnostics

Generated PDF: docs/radar/RADAR_SEARCH_PIPELINE_AS_IS.pdf

1. Purpose

This document explains the current Radar candidate-discovery pipeline as it is implemented today. It is the operational map for upstream candidate discovery, source handling, identity enrichment, diagnostic states, and benchmark evaluation.

Radar is moving from one combined "search everything" pipeline to separate pipelines. This AS IS document remains the canonical candidate-discovery AS IS until it is migrated into docs/radar/pipelines/candidate-discovery/. Future signal monitoring and Power Web discovery pipelines must have their own AS IS and TO BE Markdown/PDF documents under docs/radar/pipelines/.

The document has two jobs:

- help a user or new developer understand how Radar search currently works;
- help agents and developers find the correct extension point before changing

planner, retrieval, extraction, registry lookup, checkpoints, signal search, dossier projection, or evaluation.

The Markdown file is the source of truth. The PDF is a generated review artifact. Mermaid source can remain in Markdown, but the PDF must contain rendered diagrams instead of raw diagram notation.

2. AS IS And TO BE Rule

There is exactly one current AS IS document:

- docs/radar/RADAR_SEARCH_PIPELINE_AS_IS.md
- docs/radar/RADAR_SEARCH_PIPELINE_AS_IS.pdf

Substantial Radar pipeline changes must start with a TO BE design:

- docs/radar/to-be/RADAR_SEARCH_PIPELINE_TO_BE_<slice>.md

The TO BE document describes the intended algorithmic change before implementation. After implementation and validation, the accepted behavior is merged into this AS IS document and the PDF is regenerated.

3. Glossary

Term	Meaning
Radar definition	Persisted active configuration for one Radar, including criteria, signals, source policy, and scoring settings.
Candidate universe	The broad set of source-backed entities known to the run, including legal entities and review-needed sites, branches, projects, or assets.
Product candidate	A strict account candidate shown as a scored product row. Product candidates should be legal entities or resolved account-level entities.
Review-needed entity	A source-backed entity retained upstream with review flags, but not promoted as a confident product account.
Signal	A monitored buying-intent or activity indicator evaluated for candidate entities.
Source	A user-selected information source in Radar settings.
Connector profile	Human-readable config describing a connector, good inputs, bad inputs, expected facts, limitations, and credentials.
Capability card	Backend-compiled machine-readable connector capability used by preflight, planner input, and execution guards.
Planner source card	Compact planner-facing source description compiled from capability, source definition, and source obligation.
Source obligation	User-facing source usage mode such as required_for_identity, required_for_coverage, preferred, fallback, or disabled.
Retrieved source	URL/snippet/citation/source material returned by retrieval before extraction.
Analyzed source	Source material inspected by extraction or diagnostics but not necessarily linked to evidence.
Used source	Evidence-bearing source linked to a product candidate or finding.
Evidence ref	Stable source reference used to link extracted candidate/finding rows to normalized source records.
Checkpoint	Application-owned review point that decides whether execution continues, retries, expands, revises, stops for review, or fails hard.
Execution budget	Radar task budget for discovery, gate, signal, provider, and run-total work.
External-call budget	Budget for external actions such as OpenRouter calls, DaData lookups, provider retries, and source verification requests.
Budget reserve	A sub-allocation inside the run budget for registry identity, recall expansion, official coverage, open-web coverage, extraction recovery, or signal search.
Semantic task reserve	Application-level protected task slot for approved recall/coverage expansion tasks after the regular web-task budget is exhausted. It does not bypass external-call budgets.
Expansion target	A prioritized source-backed target that weak discovery should search next, such as a holding, legal entity, production site, branch, alias/language variant, or explicit benchmark target.
Expansion scheduler	Application-layer selector that orders guaranteed target-lane expansion probes, origin-aware completion probes, and optional expansion variants while recording scheduled/not-scheduled states.
Work scheduler	Central application-layer admission controller that decides which approved work items can consume shared budget and which must be rejected before provider execution.
Projection type loss	Diagnostic state where a source-backed review-needed entity is present, but its entity type was downgraded to unknown_entity during projection.
not_observed	A searched signal with no evidence found. It must not mean "not searched".
not_searched_*	Explicit unsearched state caused by budget, policy, missing scope, or pending output.

4. High-Level Pipeline

Figure 1. End-to-end Radar execution flow

1. Start run
API creates a queued durable Radar run.

2. Prepare runtime
Worker loads active definition and compiles source cards.

3. Plan and validate
Planner suggests; backend validates capabilities and obligations.

4. Discover and resolve
Retrieval, registry enrichment, extraction repair, entity resolution.

5. Review checkpoint
Continue, expand, repair, revise, stop, or block.

6. Signal search
Runs only after pre-signal checkpoint allows it.

7. Project and evaluate
Dossier, source lifecycle, candidate projection, benchmark metrics.

The pipeline is application-owned. Integrations execute bounded provider tasks, but they do not own scoring, review semantics, source obligations, or final candidate state.

5. Backend Roles

API route

Current owner	src/power_web_os/api
Responsibility	Create runs, expose status, candidates, dossier, runtime/preflight DTOs.
Must not own	Provider calls, scoring, SQL queries in routes.

Worker entrypoint

Current owner	src/power_web_os/jobs
Responsibility	Load run id, call application executor, persist status.
Must not own	Provider normalization or domain decisions.

Workflow wrapper

Current owner	src/power_web_os/workflows
Responsibility	Wrap application execution in workflow state when needed.
Must not own	SQLAlchemy queries, provider logic, scoring semantics.

Live run service

Current owner	src/power_web_os/application/radar/candidate_discovery/service.py
Responsibility	Order planning, staged execution, and product-safe artifact projection for one candidate-discovery run.
Must not own	Provider adapter internals, checkpoint policy, or dossier/API mapping.

Active definition adapter

Current owner	src/power_web_os/application/live_radar_definition_runtime.py
Responsibility	Convert persisted definition payload into runtime Radar payload.
Must not own	HTTP/provider details.

Connector profile registry

Current owner	src/power_web_os/application/connector_profiles.py and connector_capability_defaults.py
Responsibility	Load connector profiles and compile capability cards.
Must not own	User source obligations or provider calls.

Planner input builder

Current owner	src/power_web_os/application/live_radar_discovery_planning.py
Responsibility	Build source-card-aware planning input and deterministic fallback plans.
Must not own	Final truth or provider execution.

Planner adapter

Current owner	src/power_web_os/integrations/openrouter_discovery_planner.py
Responsibility	Ask OpenRouter for structured discovery plans.
Must not own	Source policy enforcement.

Plan acceptance

Current owner	src/power_web_os/application/live_radar_plan_acceptance.py
Responsibility	Validate capabilities, source obligations, and safe repairs.
Must not own	Provider calls.

Retrieval plan compiler

Current owner	src/power_web_os/application/live_radar_retrieval_plan.py
Responsibility	Convert accepted plan steps into bounded provider task cards.
Must not own	Scoring.

Web retrieval/extraction provider

Current owner	src/power_web_os/integrations/live_radar_openrouter.py
Responsibility	Execute OpenRouter web/retrieval/extraction tasks under budget guard.
Must not own	Execution budgets, final scoring, source obligations.

Source registry/provider orchestration

Current owner	src/power_web_os/application/radar_source_providers.py
Responsibility	Execute structured company registry providers for allowed stages.
Must not own	Signal evidence replacement.

Registry lookup term generator

Current owner	src/power_web_os/application/radar_registry_lookup_terms.py
Responsibility	Build concrete lookup terms for registry providers.
Must not own	Broad web discovery.

Search expansion service

Current owner	src/power_web_os/application/radar_search_expansion.py, radar_search_expansion_models.py, and radar_search_expansion_support.py
Responsibility	Build prioritized expansion target queues and bounded source-profile-driven query variants when discovery/coverage is weak.
Must not own	Direct provider calls.

Search expansion scheduler

Current owner	src/power_web_os/application/radar_search_expansion_scheduler.py and radar_search_expansion_selection.py
Responsibility	Select guaranteed target-lane variants, prioritize explicit benchmark completion targets over incidental targets, and order selected work before optional expansion.
Must not own	Provider calls or changing source policy.

Central work scheduler

Current owner	src/power_web_os/application/radar_work_scheduler.py
Responsibility	Admit application-approved work lanes, protect shared OpenRouter capacity for guaranteed recall expansion, and record accepted/rejected work.
Must not own	Provider calls, source policy mutation, or checkpoint decision policy.

Search expansion executor

Current owner	src/power_web_os/application/live_radar_search_expansion_execution.py
Responsibility	Execute only scheduler-admitted checkpoint expansion tasks under source policy and budget guards.
Must not own	Choosing checkpoint decisions or admitting work locally.

Extraction contract/repair

Current owner	src/power_web_os/application/live_radar_extraction_contract.py
Responsibility	Validate and repair provider payload shape when deterministic repair is safe.
Must not own	Silently converting unrecoverable output into success.

Checkpoint service

Current owner	src/power_web_os/application/radar/candidate_discovery/checkpoints/policy.py
Responsibility	Decide continue, retry, expand, repair, revise, stop, or fail.
Must not own	Direct HTTP/provider calls.

Checkpoint action executor

Current owner	src/power_web_os/application/radar/candidate_discovery/checkpoints/recovery.py
Responsibility	Apply approved checkpoint actions under budgets and policy.
Must not own	Unbounded loops.

Entity resolution

Current owner	src/power_web_os/application/live_radar_entity_resolution.py
Responsibility	Distinguish legal entity, branch, production site, project, asset, and unknown entity.
Must not own	Provider transport.

Candidate universe support

Current owner	src/power_web_os/application/live_radar_universe.py and retrieved-candidate helpers
Responsibility	Preserve source-backed legal entities and review-needed upstream entities.
Must not own	Product precision claims.

Live artifact projection

Current owner	src/power_web_os/application/radar/candidate_discovery/diagnostics/live_run_artifact.py
Responsibility	Shape product-safe live run artifacts from completed application state.
Must not own	Provider execution, checkpoint decisions, or scheduler admission.

Dossier projection

Current owner	src/power_web_os/api/radar_dossier_mappers.py and related mappers
Responsibility	Explain lifecycle, diagnostics, checkpoints, budgets, candidates, and sources.
Must not own	Mutating run behavior.

Evaluation

Current owner	src/power_web_os/radar_evaluation.py
Responsibility	Compare persisted run/dossier output to curated baseline.
Must not own	Live provider calls.

6. Inputs And Runtime Configuration

The live run is driven by these inputs:

- active RadarDefinitionRecord loaded by radar_id;
- canonical runtime Radar payload built from the active definition;
- global_search_policy.sources;
- source usage obligations from Radar settings;
- connector profiles under config/connectors;
- compiled capability cards and planner source cards;
- runtime config from .env and task context;
- execution budgets and external-call budgets;
- run profile such as live, smoke, benchmark_smoke, or benchmark_live;
- persisted run metadata from API queue time and worker execution time.

Secrets stay in .env or deployment secret storage. They are never passed to planner source cards, dossier, journal, evaluation reports, or AS IS/TO BE docs.

7. Planning Loop

Figure 2. Planner, source cards, and backend validation

Current planning behavior:

1. The worker loads the active persisted Radar definition. The hardcoded mini Radar is legacy/offline fallback only.
2. Connector profiles are compiled into source cards for configured sources.
3. Planner input includes source cards and source obligations. Source cards say what a source can do; obligations say how strongly the selected source must be used.
4. The planner may use OpenRouter and must return a structured plan.
5. Backend validation rejects incompatible source use, disabled sources, missing required source use, and lookup-only sources used for broad discovery.
6. Invalid plans can enter a bounded revision loop. If revision remains invalid, execution stops as review-needed or policy-blocked instead of falling into blind execution.

Planner calls count against OpenRouter external-call budgets, including total OpenRouter run budget and planner-role budget.

8. Source Cards And Connector Profiles

Connector profiles are human-facing config. They describe a connector in terms a connector author can understand:

- display name;
- description;

- good inputs;
- bad inputs;
- expected facts;
- limitations;
- credential environment variable names;
- runtime provider id.

Application code compiles those profiles into capability cards:

- lookup-only or broad-discovery-capable;
- identity/enrichment/coverage/signal applicability;
- required input kinds;
- returned fact kinds;
- useful-result criteria;
- accepted input shapes;
- bad input shapes;
- non-blocking outcomes;
- language/alias hints;
- connector capability class;
- credential requirements.

Planner source cards are compact, product-safe versions of the capabilities.

They contain no credentials, API keys, headers, tokens, or provider secrets.

9. Retrieval, Extraction, And Recovery Loop

Pipeline-critical LLM calls must return structured JSON. The shared contract is:

1. Call the primary role model.
2. Parse JSON and validate the role-specific application schema.
3. Retry the primary model once with strict repair context when output is

non-JSON or schema-invalid.

4. Retry the configured backup model once when primary retry is still invalid.
5. Stop the affected branch with an exact diagnostic reason when recovery is

exhausted.

Current role routing:

Planner

Primary model setting	OPENROUTER_PLANNER_MODEL
Backup model setting	OPENROUTER_PLANNER_BACKUP_MODEL, then OPENROUTER_BACKUP_MODEL
Temperature setting	OPENROUTER_PLANNER_TEMPERATURE

Extraction

Primary model setting	OPENROUTER_EXTRACTOR_MODEL
Backup model setting	OPENROUTER_EXTRACTION_BACKUP_MODEL, then OPENROUTER_BACKUP_MODEL
Temperature setting	OPENROUTER_EXTRACTOR_TEMPERATURE

Signal/default

Primary model setting	OPENROUTER_MODEL
Backup model setting	future role-specific backup
Temperature setting	OPENROUTER_SIGNAL_TEMPERATURE

Backup attempt

Primary model setting	configured role backup
Backup model setting	none
Temperature setting	OPENROUTER_BACKUP_TEMPERATURE

Default temperature is 0. Retry and backup attempts count against both external OpenRouter budgets and provider-retry budgets. Technical trace records attempt role, attempt index, model, temperature, and failure reason without headers, API keys, hidden reasoning, or raw provider dumps.

Retrieval And Extraction Path

The retrieval/extraction path is split conceptually:

1. Retrieval obtains ranked sources, snippets, URLs, citations, and source outcomes.
2. Extraction maps task cards plus retrieved material into structured observations, candidates, source outcomes, coverage findings, and candidate universe gaps.
3. Schema validation checks provider output shape.
4. Deterministic repair attempts safe shape fixes.
5. If repair fails, one bounded primary retry can be attempted with strict JSON context.
6. If configured, one bounded backup extraction model retry can be attempted for extraction-stage tasks only.
7. If still invalid, the branch stops with an explicit diagnostic reason such as primary_schema_invalid, backup_schema_invalid, backup_not_configured, or budget_exhausted_before_backup.

Extraction recovery attempts are recorded in extraction_recovery_records and must count against external-call and provider-retry budgets.

10. Registry Lookup Loop

Company registry providers are structured identity/enrichment sources. DaData is the first implementation. Registry providers are not broad discovery engines and must not replace web evidence for signals.

The current lookup flow:

1. Build concrete lookup terms from candidate names, retrieved snippets, identifiers, Russian aliases, English aliases, legal-form variants, and short fragments.
2. Reject placeholders such as "candidates from step 1" and broad natural

language discovery queries.

3. Try terms in bounded order: identifiers, Russian legal-form terms, Russian short names, then English aliases.
4. Count each lookup term against DaData lookup budget.
5. Stop after a useful match.
6. Preserve each lookup attempt in registry_lookup_attempts.

Important semantics:

- one alias no_match is not a hard block;
- registry_lookup_insufficient means there was no concrete registry input;
- identity_not_confirmed_after_all_terms means concrete terms were tried but identity still was not confirmed;
- ambiguous but source-backed entities can be retained as review-needed upstream entities or linked facts.

11. Search Expansion Loop

Figure 3. Checkpoint decision and recovery loop

Checkpoint input
candidate universe, source counts, obligations, extraction/linking issues, budget state

Backend decision
continue, recover, stop for review, or fail hard

Recovery actions are mutually exclusive and bounded:

Retry same source
same bounded task

Expand search
allowed official/open web

Repair extraction
deterministic repair or bounded retry

Revise plan
only plan/policy strategy failure

Continue
next stage or signal search

Stop / block
explicit diagnostic terminal state

Loop rule
Recovery is budgeted, policy-checked, merged, and re-reviewed. No blind continuation.

When discovery or coverage is weak, RadarSearchExpansionService first builds a prioritized expansion target queue, then compiles bounded query variants for the highest-priority targets.

Checkpoint recovery now uses the same target-aware expansion path. If the run has weak recall, low linked-source coverage, explicit benchmark target hints, or source-backed unresolved gaps, the checkpoint can choose expand_sources before asking for a plan revision. This prevents evidence_linking_failed from automatically consuming the revision cap when the real problem is that important targets were not retrieved yet.

Target classes:

- holding_or_group_target;
- production_site_or_branch_target;
- known_subsidiary_or_legal_entity_target;
- source_backed_universe_gap_target;
- alias_or_language_variant_target;
- benchmark_baseline_like_target only when explicit benchmark context exists;
- low_confidence_registry_suggestion_target.

Each target records target_id, target_label, target_type, source_refs, why_target_exists, priority, allowed source ids, expected fact kinds, budget_reserve_key, execution status, and not-searched reason. It also records selection context such as target_origin, completion_rank_reason, deprioritized_reason, and uncovered_baseline_target. These fields explain whether the target came from explicit benchmark context, retrieved evidence, candidate gaps, aliases, or a generic seed.

Variant selection is target-aware and guarantee-aware. The planner no longer takes a flat top-N list of query variants. It first deduplicates variants, then selects benchmark target-lane minimums before optional variants. In benchmark_smoke, the effective variant cap is raised to at least the sum of required target lanes, so a low generic cap cannot make the guarantee impossible before scheduler admission.

When benchmark target minimums are present, selection first tries to choose one holding/group target, two legal/subsidiary targets, and two production-site/branch targets. Optional alias, source-gap, or extra target variants are added only after those minimums are selected. Production-site and branch targets remain lane-prioritized because they drive review recall, while product candidate projection remains strict.

Targets that are generated but do not receive an execution slot are kept in targets_not_searched with a specific reason. Selection-level reasons include target_not_generated, no_executable_variant_for_target, and selection_below_minimum; scheduler/admission reasons remain budget-specific.

After target-lane ordering, RadarWorkScheduler performs central admission. This is the boundary between "selected" and "allowed to spend provider budget". For each scheduled expansion task it records a RadarWorkItem, RadarWorkAdmissionDecision, lane summary, execution order, rejected work, and guarantee failures. A rejected guaranteed work item is visible before provider execution as work_admission_rejected; it is not counted as searched.

Query variants are compiled from connector source cards and policy:

- official/domain-capable sources produce site:<domain> <target> plus relation and industrial coverage queries;
- broad web sources produce relation, membership, identity, and industrial/site queries;
- lookup-only registry sources are not used for broad expansion tasks;
- source-specific domains come from the selected source/profile, not a hardcoded production branch;
- relation terms come from the radar definition and target context.

Expansion respects source policy and budget reserves. If open web is disabled, open-web variants are not generated. If an official source is disabled, official-domain variants are not generated. If a reserve is exhausted, the target/task is recorded in targets_not_searched with not_searched_reason instead of disappearing from diagnostics.

Production-site, branch, asset, and project targets use the dedicated production_site_coverage_probe reserve. In smoke profile this reserve defaults to 2; benchmark_smoke overrides it to 3 so a bounded SIBUR contour smoke can attempt the three curated production-site misses before broader live benchmarking.

Budget reserves are not just labels. Expansion tasks registered under a recall-expansion reserve can use two protected layers:

- an application-level semantic task reserve if the regular Radar web-task budget is exhausted;
- a protected openrouter_recall_expansion external-call slot if the regular openrouter_web_task role budget is exhausted.

Before a scheduled guaranteed expansion task reaches the provider, the executor asks RadarWorkScheduler for admission. The scheduler checks total OpenRouter, protected recall-expansion OpenRouter, OpenRouter server-tool web-search, and budget-reserve capacity before the provider is called. If capacity is already gone, the target is recorded as a rejected work item with a reason such as external_total_budget_limited, openrouter_recall_expansion_budget_limited, server_tool_budget_limited, or budget_reserve_exhausted.

Guaranteed recall-expansion work has one additional protection: every accepted guaranteed expansion task receives a reserved first OpenRouter recall-expansion call. Retries and optional work may use only headroom after those first calls are protected. If a retry would spend the last call needed by another accepted guaranteed task, it is blocked with guaranteed_external_reservation_protected. If the scheduler cannot reserve enough external capacity for guaranteed work before provider execution, the admission reason is guaranteed_external_reservation_insufficient.

Both semantic and external layers must pass before the provider call is made. Semantic task reserves do not bypass total OpenRouter calls, server-tool web-search calls, source verification, source policy, or connector capability guards. In benchmark_smoke, regular web-task calls are capped at 10, protected recall-expansion OpenRouter calls are capped at 7, total OpenRouter calls are capped at 20, and server-tool web-search calls are capped at 60.

Benchmark smoke also carries target-lane guarantees in task context. The current minimums are one holding/group probe, two legal/subsidiary probes, and two production-site/branch probes. The guarantee is evaluated from executed expansion results, not from generated target queues. If the minimum is not met, target_probe_guarantee_failures names the blocker, such as target_not_generated, no_executable_variant_for_target, selection_below_minimum, target_not_selected, scheduled_below_minimum, semantic_task_budget_limited, external_budget_limited, source_policy_limited, or executed_below_minimum.

After the lane minimums are selected, benchmark_smoke can run a bounded coverage-completion selection pass. This pass does not call providers and does not bypass the scheduler. It simply adds a small number of still-uncovered generated targets to the selected variant set before work admission. The current benchmark_smoke completion limit is coverage_completion_target_limit=2. Completion targets are chosen after mandatory lane minimums, so uncovered targets do not steal the base holding/legal/site minimums. Inside the completion pass, uncovered benchmark/baseline targets are ranked before ordinary targets; then target type, target origin, label quality, coverage novelty, and variant reason break ties. This preserves the previous production-site completion behavior while giving remaining legal/subsidiary misses a fair slot after the mandatory lanes are satisfied. If the completion pass still cannot select a target, diagnostics use specific reasons such as completion_cap_exhausted or selector_priority_lost rather than a blank not_retrieved_in_run, and include the rank reason that explains why the target lost.

Expansion diagnostics include expansion_target_summary_by_type, search_expansion_selection_summary, search_expansion_selection_diagnostics, expansion_schedule, target_lane_allocation, search_expansion_results_by_target_type, search_expansion_execution_summary, search_expansion_target_coverage, targets_not_searched, semantic task reserve counters, and external reserve counters. After 0.7.6.3.6.7, diagnostics also include work_scheduler_plan, work_scheduler_ledger, work_admission_decisions, work_lane_summary, work_guarantee_failures, work_execution_order, deferred_work_items, and rejected_work_items. Scheduler diagnostics are aggregated across all expansion portfolios in a run. Later checkpoints must not overwrite earlier admissions, because that would hide which guaranteed lanes were actually admitted before provider spending.

After 0.7.6.3.6.13, dossier and benchmark reports also include legal_subsidiary_completion_summary. It is a compact funnel for known_subsidiary_or_legal_entity_target: generated targets, selected variants, executed targets, not-searched targets, and reason buckets. It exists so a SIBUR-like benchmark can tell whether a legal/subsidiary miss was never generated, lost the completion cap, was rejected by scheduler admission, hit external budget, or found a source that was not projected.

The execution summary is a funnel: generated targets, selected variants, attempted tasks, externally executed provider calls, sources found, and projected entities. A target with budget_decision.accepted=false is not counted as searched. Evaluation can therefore distinguish expansion_not_selected, completion_cap_exhausted, selector_priority_lost, completion_not_selected, expansion_global_budget_limited, expansion_reserve_limited, external_budget_limited, scheduler_rejected,

semantic_task_budget_limited,
expansion_searched_no_support, expansion_source_found_not_projected, and present_not_projected instead of reporting every production-site miss as a blank not_retrieved_in_run.

For benchmark runs only, task_context.benchmark_target_hints can seed benchmark_baseline_like_target expansion targets. Production runs ignore those hints unless an explicit benchmark_profile is present.

12. Candidate Universe And Entity Resolution

The upstream candidate universe is recall-first. It can retain source-backed entities that are not yet strict product accounts.

Current rules:

- legal entities can become product candidates when evidence and resolution are sufficient;
- weak legal-entity mentions can remain review-needed universe entries;
- branches, factories, production sites, projects, and assets can be retained as review-needed upstream entities or linked facts;
- unresolved sites, projects, or assets must not become high-confidence product account candidates;
- review flags explain why an entity needs human attention;
- typed upstream entities can upgrade duplicate unknown_entity universe rows instead of being skipped as duplicates.

Common review flags include:

- requires_human_review;
- not_standalone_legal_entity;
- registry_match_ambiguous;
- official_source_cross_checked;
- candidate_universe_gap.

Product candidate projection remains precision-first. Upstream universe retention is allowed to be broader than product account output.

Projection must preserve review-needed metadata across handoffs:

- entity_type;
- resolution_status;
- resolved_legal_name;
- not_candidate_reason;
- review_flags;
- source_refs.

If a candidate-universe gap enters first without type metadata and a typed upstream disambiguation record arrives later with the same name, the existing universe row is upgraded in place. This prevents source-backed production sites from degrading to unknown_entity before evaluation.

13. Checkpoints And Adaptive Actions

The checkpoint service reviews execution after key stages:

- after discovery;
- after qualification gates;
- after coverage checks;
- before signal search.

Possible decisions:

- continue;
- retry_same_source;
- expand_sources or search expansion;
- repair_extraction;
- retry_extraction;
- revise_plan;
- stop_review_needed;
- fail_hard.

Actions are bounded:

- no unbounded retries;
- no policy bypass;
- no hidden broad fallback;
- weak recall with uncovered target hints runs target-aware expansion before revision;
- repeated unlinked evidence after expansion can still route to revise_plan;
- no signal search until the pre-signal checkpoint allows it;
- all adaptive provider calls count against budgets.

If a stop is diagnostic rather than a runtime failure, the run can still produce a completed snapshot with stopped_for_review_reason and explicit checkpoint metadata.

14. Signal Search

Signal search is not candidate discovery. It runs only after candidate universe and qualification/coverage checkpoints allow it.

Rules:

- signal tasks should use web/official evidence sources, not registry enrichment sources unless a connector capability explicitly supports signal evidence;
- signal prompts include one signal and the current candidate scope;
- signal extraction must not add new scored candidates;
- new entities found during signal search go to candidate universe gaps or diagnostics;
- unsearched signals become not_searched_*, not not_observed;
- not_observed is reserved for searched-negative evidence or invalid searched

evidence.

15. Budget Model

Radar has two complementary budget layers.

Execution budgets:

- discovery tasks per rule;
- gate tasks per candidate/rule;
- signal tasks per candidate/signal;
- provider task keys;
- total web tasks per run.

External-call budgets:

- total OpenRouter HTTP calls;
- OpenRouter planner calls;
- OpenRouter web task calls;
- protected OpenRouter recall-expansion calls;
- OpenRouter server-tool web searches reported after the call;
- DaData/company registry lookups;
- source verification requests;
- provider retries per task.

Smoke and benchmark profiles set bounded task context budgets so acceptance runs do not spread into uncontrolled provider calls. OpenRouter latency alone is not an error; the controllable unit is number of external actions.

Budget reserves add an intent-level layer under the same run:

- primary_discovery;
- registry_identity;
- recall_expansion;
- official_coverage_probe;
- open_web_coverage_probe;
- production_site_coverage_probe;
- extraction_recovery;
- signal_search.

Reserve counters are reported as budget_reserve_counters. Exhausted reserves are reported as budget_reserve_exhaustion_events and should produce explicit not-searched diagnostics for skipped expansion targets or registry identity attempts. Protected recall-expansion calls are reported in external counters as openrouter_recall_expansion:run, while still incrementing the total openrouter:run counter.

RadarWorkScheduler can reserve part of the total OpenRouter run capacity for guaranteed recall-expansion lanes. Regular OpenRouter web-task calls are rejected with work_admission_reserved_capacity when accepting them would consume the shared capacity needed for admitted recall expansion. Protected recall-expansion calls still count against openrouter:run; the reservation only prevents earlier optional work from spending the last shared slots.

For guaranteed recall-expansion work, the reservation is tracked per accepted task. Dossier and benchmark reports expose work_admission_reserved_capacity.guaranteed_recall_expansion, including reserved task count, first calls used, and first calls still remaining.

Semantic task reserves are separate from external-call reserves. They live in RadarExecutionBudget and are configured through task_context.semantic_task_reserve_limits. In benchmark_smoke, the current defaults are:

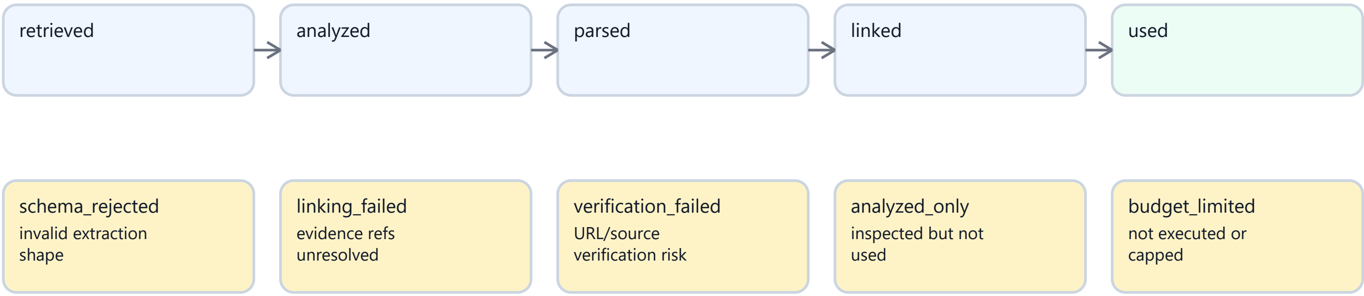
- recall_expansion: 6;
- production_site_coverage_probe: 3;
- official_coverage_probe: 3;
- open_web_coverage_probe: 3.

They allow approved expansion tasks to execute after the regular web-task budget is exhausted. They still increment visible total task counters and are reported as semantic_task_budget_counters and semantic_task_budget_exhaustion_events.

Source verification uses a per-run URL cache. The cache key lowercases the host, strips URL fragments, and normalizes trailing slashes. Duplicate URL checks reuse the cached reachability result and do not spend another source_verification budget unit. Dossier and benchmark reports expose source_verification_cache_stats, source_verification_unique_request_count, and source_verification_duplicate_skip_count.

16. Source Lifecycle

Figure 4. Source lifecycle and rejection branches



Product source list includes only 'used'. Dossier lifecycle keeps all diagnostic states.

Product source lists remain strict: only evidence-bearing used sources appear as product sources. Dossier/source lifecycle diagnostics are broader and must show retrieved, analyzed, rejected, unlinked, verification-limited, and budget-limited sources.

17. Dossier, Journal, Trace, And Evaluation

Radar exposes several diagnostic surfaces:

- run status and run metadata;
- dossier summary and source lifecycle;
- candidate universe and product candidates;
- checkpoint decisions and adaptive actions;
- source cards and capability validation;
- source obligation decisions and runtime outcomes;
- external-call budget counters;
- extraction recovery records;
- registry lookup terms and attempts;
- expansion target queue;
- search expansion query variants and results grouped by target;
- targets not searched and budget reserve counters;
- semantic task budget counters and target probe guarantees;
- source verification cache statistics;
- registry ambiguity fan-out summary;
- technical trace for sanitized developer inspection;
- benchmark and evaluation reports.

Forbidden everywhere:

- API keys;
- authorization headers;
- bearer tokens;

- raw prompts when not explicitly sanitized;
- raw hidden chain-of-thought;
- raw provider dumps that contain secrets or hidden reasoning keys.

18. Evaluation Loop

The benchmark/evaluation layer reads persisted run output and dossier data. It does not call OpenRouter, DaData, or source providers.

Current SIBUR evaluation channels:

- strict_recall for legal-entity baseline hits;
- review_recall for production-site, branch, asset, or project hits retained

as review-needed universe entities or linked facts;

- precision for strict product account candidates;
- false positives;
- false negatives;
- ambiguous matches;
- evidence quality buckets;
- optional coverage probe output that is diagnostic only and does not change the

original metrics.

Review recall can match non-legal baseline entities through product-safe diagnostic channels:

- candidate_universe;
- upstream_disambiguation_results;
- linked_entity_facts;
- unresolved_candidate_gaps.

The matcher is source-backed and baseline-driven. Exact normalized names remain strongest, while production-site/branch review matches can tolerate generic relation suffixes such as parenthesized department names when the meaningful baseline tokens are present. unknown_entity rows do not count as review hits; they become projection_type_lost when the entity text is present but the review entity type was lost.

Evaluation is a measurement layer. If it exposes poor quality, the fix should be planned as a follow-up slice rather than hidden inside the evaluation code.

19. Context Management

Figure 5. Context boundaries across pipeline roles

Inputs
Active definition, Source policy, Connector profiles, Runtime budgets

Application decisions
Planner cards, Task cards, Observations, Checkpoints

Outputs
Dossier, Candidate universe, Product candidates, Evaluation

Forbidden context
Never pass secrets, raw hidden reasoning, headers, tokens, or raw provider dumps.

Planner

Allowed context	Radar goal, source cards, obligations, compact criteria/signals, budget hints.
Forbidden context	Secrets, raw provider dumps, hidden reasoning, credentials.

Plan reviser

Allowed context	Product-safe checkpoint facts and capability validation errors.
Forbidden context	Raw traces, hidden reasoning, secret-bearing payloads.

Web extractor

Allowed context	Task card, retrieved sources, expected schema, current candidate scope.
Forbidden context	Full unrelated run history, secrets, scoring decisions.

Backup extractor

Allowed context	Failed extraction reason and strict task-specific JSON context.
Forbidden context	Planner reasoning, broad source policy mutation.

Registry provider

Allowed context	Concrete legal names, INN, OGRN, legal-form variants, strong aliases.
Forbidden context	Broad discovery queries and placeholders.

Checkpoint service

Allowed context	Counts, outcomes, warnings, budgets, source obligations, extraction issues.
Forbidden context	Direct provider credentials or raw hidden reasoning.

Entity resolver

Allowed context	Provider observations, registry facts, source refs, task context.
Forbidden context	HTTP client details.

Dossier mapper

Allowed context	Sanitized execution metadata and output snapshot.
Forbidden context	Runtime side effects or scoring changes.

Evaluation

Allowed context	Persisted dossier/output and baseline fixture.
Forbidden context	Live provider calls.

20. Extension Points

New source connector

Extension point	config/connectors, connector profile registry, source provider port/adapter.
Required validation	Connector profile tests, preflight, source card validation.

Planner capability change

Extension point	source card compiler, planning input, plan acceptance.
Required validation	Planner/validator fake tests, no-secret trace tests.

Structured LLM role or model routing

Extension point	role adapter, OpenRouter request builder, runtime config, universal LLM call contract ADR.
Required validation	Non-JSON test, schema-invalid test, retry/backup budget test, no-secret trace test.

Search expansion strategy

Extension point	RadarSearchExpansionService.
Required validation	Unit tests for query families, policy filtering, caps, dedupe.

Registry lookup terms

Extension point	RegistryLookupTermGenerator.
Required validation	Unit tests for aliases, Russian/English/legal-form terms, identifiers, placeholders.

Extraction schema repair

Extension point	extraction contract and OpenRouter provider recovery path.
Required validation	Malformed-output fixtures, retry/backup budget tests.

Entity resolution semantics

Extension point	entity resolution and candidate universe projection.
Required validation	Legal entity vs site/branch/project/asset tests.

Checkpoint policy

Extension point	checkpoint service/action executor.
Required validation	Recorded/fake adaptive pipeline tests.

Signal search behavior

Extension point	retrieval task compiler and staged execution signal phase.
Required validation	Not-searched vs not-observed tests.

Dossier projection

Extension point	API dossier mappers and source lifecycle.
Required validation	Backend API/dossier tests.

Benchmark metrics

Extension point	evaluation module and baseline fixtures.
Required validation	Evaluation unit/report tests.

21. Test Map

Behavior	Typical tests
Active definition and runtime wiring	tests/test_radar_preflight.py, tests/test_persisted_live_radar.py
Connector profiles and source cards	tests/test_connector_profiles.py, tests/test_live_icp_radar.py
Source obligations	tests/test_live_icp_radar.py, tests/test_radar_preflight.py
Execution, semantic, and external-call budgets	tests/test_radar_external_call_budget.py

Behavior	Typical tests
Central work scheduler and admission control	tests/test_radar_work_scheduler.py
Adaptive checkpoints	tests/test_radar_adaptive_execution.py
Search expansion and lookup terms	tests/test_radar_search_expansion.py, tests/test_live_icp_radar.py
Extraction recovery	tests/test_live_icp_radar.py, tests/test_radar_adaptive_execution.py
Dossier/API projection	tests/test_backend_api.py
Benchmark report	tests/test_radar_benchmark.py
Recall/precision evaluation	tests/test_radar_evaluation.py
Backend boundaries	tests/test_backend_architecture_contract.py

22. AS IS / TO BE Maintenance Lifecycle

Figure 6. AS IS / TO BE maintenance cycle

1 AS IS
current implementation

2 TO BE
planned pipeline change

3 Review
user and developer alignment

4 Implement
slice code and tests

5 Validate
fixtures, smoke, evaluation

6 Finalize

Rule
A substantial Radar pipeline slice is Done only after implemented behavior is reflected in AS IS Markdown/PDF.

Required maintenance rules:

1. Before a substantial Radar pipeline change, create a TO BE document.
2. Generate or update the slice plan from that TO BE.
3. Implement the slice.
4. Run targeted tests and any required smoke/evaluation flow.
5. Update AS IS with implemented behavior.
6. Regenerate PDF.
7. Mark the roadmap slice Done only after AS IS/PDF are current.

23. Known Gaps

- This document is descriptive. It does not replace tests, dossier, trace, or benchmark reports.
- The AS IS document can drift if future slices skip the maintenance rule; the documentation contract test and slice acceptance criteria exist to reduce that risk.
- The PDF rendering command is intentionally project-local and may need tooling adjustment if the repository moves to another documentation stack.